

Polymers

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

Q1 JEE Main 2020 - 9 January (Evening)

Which polymer has chiral monomer(s)?

- (A) Buna-*N*
- (B) PHBV
- (C) Neoprene
- (D) Nylon 6, 6

Q2 JEE Main 2020 - 5 September (Evening)

Which one of the following polymers is not obtained by condensation polymerisation?

- (A) Buna-N
- (B) Nylon 6
- (C) Bakelite
- (D) Nylon 6,6

Q3 JEE Main 2020 - 6 September (Morning)

Consider the Assertion and Reason given below.

Assertion (A) : Ethene polymerized in the presence of Ziegler Natta Catalyst at high temperature and pressure is used to make buckets and dustbins.

Reason (R) : High density polymers are closely packed and are chemically inert.

Choose the correct answer from the following:

- (A) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (B) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (C) (A) is correct but (R) is wrong
- (D) (A) and (R) both are wrong

Q4 JEE Main 2020 - 6 September (Evening)

The correct match between Item - I and Item - II is

Item - I**(a) Natural rubber****(b) Neoprene****(c) Buna-N****(d) Buna-S****Item - II****(I) 1, 3-butadiene
+ styrene****(II) 1, 3-butadiene
+ acrylonitrile****(III) Chloroprene****(IV) Isoprene**

(A) (a) – (111), (b) – (IV), (c) – (1), (d) – (11)

(B) (a) – (IV), (b) – (III), (c) – (II), (d) – (1)

(C) (a) – (III), (b) – (IV), (c) – (II), (d) – (1)

(D) (a) – (IV), (b) – (III), (c) – (I), (d) – (II)

Q5 JEE Main 2020 - 8 January (Evening)

Preparation of Bakelite proceeds via reactions :

(A) Electrophilic substitution followed by condensation.

(B) Nucleophilic addition followed by dehydration.

(C) Electrophilic addition followed by dehydration.

(D) Hydration followed by condensation.

Polymers

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (B)

Q2 (A)

Q3 (A)

Q4 (B)

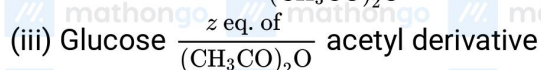
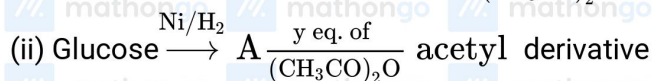
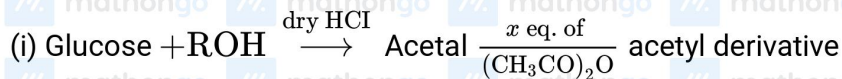
Q5 (A)

#MathBoleTohMathonGo

www.mathongo.com

Q1 JEE Main 2020 - 2 September (Morning)

Consider the following reactions:



'x', 'y' and 'z' in these reactions are respectively.

(A) 4, 5 & 5

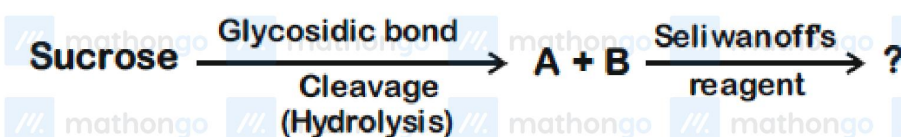
(B) 4, 6 & 5

(C) 5, 4 & 5

(D) 5, 6 & 5

Q2 JEE Main 2020 - 2 September (Evening)

The correct observation in the following reactions is :



(A) Gives no colour

(B) Formation of red colour

(C) Formation of violet colour

(D) Formation of blue colour

Q3 JEE Main 2020 - 3 September (Evening)



The number of  groups present in a tripeptide Asp - Glu - Lys is Answer (5) Sol. The amino acids present in the given tripeptide

Q4 JEE Main 2020 - 4 September (Morning)

What are the functional groups present in the structure of maltose?

Biomolecules

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

- (A) One acetal and one ketal
- (B) One ketal and one hemiketal
- (C) Two acetals
- (D) One acetal and one hemiacetal

Q5 JEE Main 2020 - 4 September (Evening)

The number of chiral centres present in threonine is

Q6 JEE Main 2020 - 5 September (Morning)

Which of the following is not an essential amino acid?

- (A) Tyrosine
- (B) Valine
- (C) Lysine
- (D) Leucine

Q7 JEE Main 2020 - 5 September (Evening)

The number of chiral carbons present in sucrose is _____.

Q8 JEE Main 2020 - 6 September (Evening)

Which one of the following statements is not true?

- (A) Lactose contains α -glycosidic linkage between C_1 of galactose and C_4 of glucose
- (B) Lactose is a reducing sugar and it gives Fehling's test
- (C) On acid hydrolysis, lactose gives one molecule of D(+)-glucose and one molecule of D(+)-galactose
- (D) Lactose ($C_{11}H_{22}O_{11}$) is a disaccharide and it contains 8 hydroxyl groups

Q9 JEE Main 2020 - 7 January (Morning)

Biomolecules

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Column-I

- (A) Thiamine
- (B) Riboflavin
- (C) Pyridoxine
- (D) Ascorbic acid

Match the column

- (A) $A \rightarrow Q, B \rightarrow R, C \rightarrow P, D \rightarrow S$
- (B) $A \rightarrow Q, B \rightarrow R, C \rightarrow S, D \rightarrow P$
- (C) $A \rightarrow R, B \rightarrow Q, C \rightarrow S, D \rightarrow P$
- (D) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$

Column-II

- (P) Scurvy
- (Q) Beri Beri
- (R) Cheilosis
- (S) Convulsions

Q10 JEE Main 2020 - 7 January (Evening)

The correct statement about gluconic acid is

- (A) It is prepared by oxidation of glucose with HNO_3
- (B) It is obtained by partial oxidation of glucose
- (C) It is dicarboxylic acid
- (D) It forms hemiacetal or acetal

Q11 JEE Main 2020 - 8 January (Morning)

Number of chiral centres in Pencillin is

Q12 JEE Main 2020 - 8 January (Morning)

Which of the given statement is incorrect about glucose?

- (A) Glucose exists in two crystalline forms α and β
- (B) Glucose gives Schiff's test for aldehyde.
- (C) Glucose reacts with hydroxylamine to form oxime.
- (D) The pentaacetate of glucose does not react with hydroxylamine to give oxime.

Q13 JEE Main 2020 - 8 January (Evening)

Two monomers in maltose are :

- (A) α -D-Glucose, α -D-Glucose
- (B) α -D-Glucose, β -D-Glucose

Biomolecules

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

(C) α -D-Galactose, β -D-Glucose

(D) β -D-Galactose, α -D-Glucose

Q14 JEE Main 2020 - 9 January (Evening)

A , B and C are three biomolecules. The results of the tests performed on them are given below:

A , B and C are respectively:

	Molisch's Test	Barfoed Test	Biuret Test
A	Positive	Negative	Negative
B	Positive	Positive	Negative
C	Negative	Negative	Positive

(A) A = Lactose, B = Glucose, C = Alanine

(B) A = Glucose, B = Fructose, C = Albumin

(C) A = Lactose, B = Glucose, C = Albumin

(D) A = Lactose, B = Fructose, C = Alanine

Q15 JEE Main 2020 - 4 September (Morning)

Which of the following will react with $\text{CHCl}_3 + \text{alc. KOH}$?

(A) Adenine and thymine

(B) Thymine and proline

(C) Adenine and lysine

(D) Adenine and proline

Biomolecules

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (D)

Q2 (B)

Q3 (5)

Q4 (D)

Q5 (2)

Q6 (A)

Q7 (9)

Q8 (A)

Q9 (B)

Q10 (B)

Q11 (3)

Q12 (B)

Q13 (A)

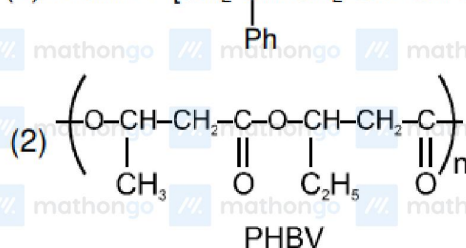
Q14 (C)

Q15 (C)

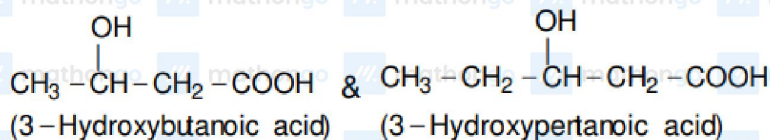
#MathBoleTohMathonGo

www.mathongo.com

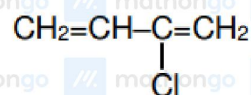
Q1

Polymers(1) Buna-S $-\text{[CH}_2-\text{CH}(\text{Ph})-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2]-$ 

In PHBV, both monomers have chiral centre.

(3) Neoprene $-\text{[CH}_2-\text{CH}=\text{CCl}-\text{CH}_2]_n-$ **Monomers** $\text{CH}_2=\text{CH}-\text{Ph}$ & $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ 

(4) Nylon-6,6

 $\sim[\text{CO}(\text{CH}_2)_4\text{CO}-\text{NH}(\text{CH}_2)_6\text{NH}]_n\sim$  $\text{HO}_2\text{C}-(\text{CH}_2)_4-\text{CO}_2\text{H}$ & $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$

Q2

Buna-N : Obtained by addition polymerisation Nylon 6, Bakelite, Nylon 6,6 : Obtained by condensation polymerisation.

Q3

High density polyethene is hard and chemically inert that's why used to make buckets and dustbins.

Q4

Natural rubber - Polymer of isoprene

Neoprene

- Polymer of chloroprene

Buna N

- Polymer of 1,3-Butadiene and acrylonitrile

Buna S

- Polymer of 1,3-Butadiene and styrene

Q5

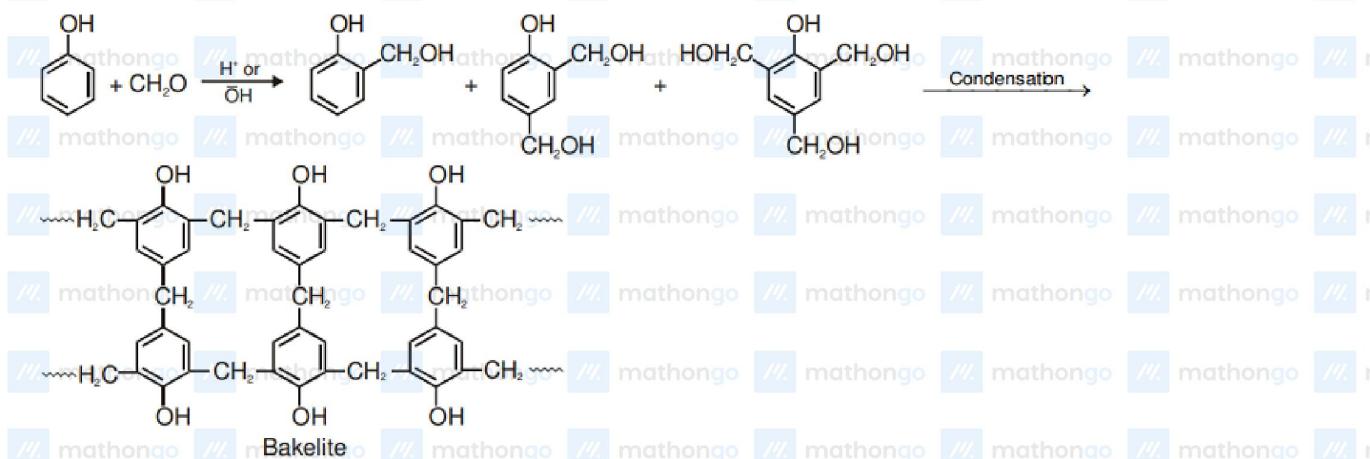
Polymers

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Formation of Bakelite follows electrophilic substitution reaction of phenol with formaldehyde followed by condensation.

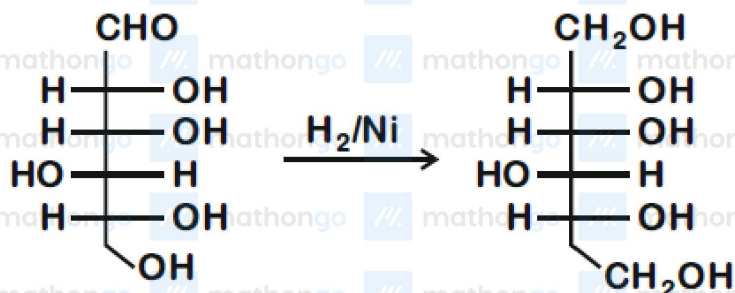


Q1

5 eq. of $(\text{CH}_3\text{CO})_2\text{O}$ is used in (i) reaction.

6 eq. of $(\text{CH}_3\text{CO})_2\text{O}$ is used in (ii) reaction

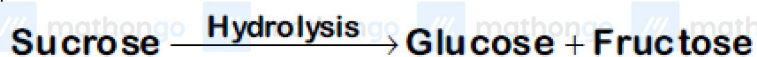
5 eq. of $(\text{CH}_3\text{CO})_2\text{O}$ is used in (iii) reaction



5 eq. of $(\text{CH}_3\text{CO})_2\text{O}$
will be used in this
case

6 eq. of $(\text{CH}_3\text{CO})_2\text{O}$
will be used in this
case

Q2



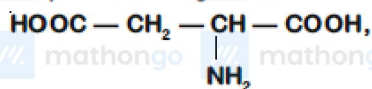
Seliwanoff's
reagent

Red colour will
be observed

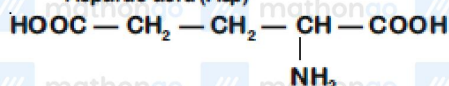
Ketose with seliwanoff's reagent gives red colour. It is a specific test for ketose.

Q3

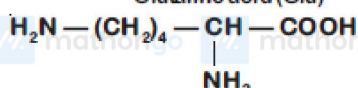
Asp – Glu – Lys are



Aspartic acid (Asp)



Glutamic acid (Glu)

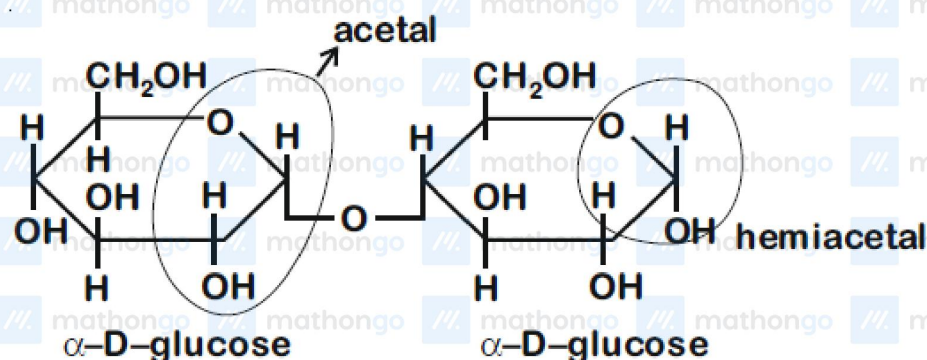


Lysine (Lys)

$\therefore$ Number of carbonyl groups present in tripeptide = $2 + 2 + 1 = 5$

Q4

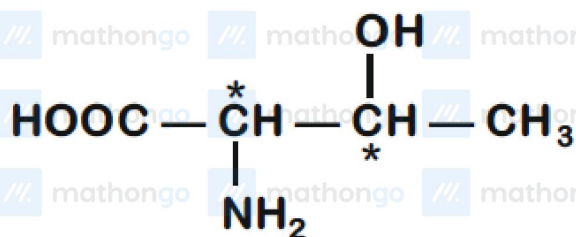
Maltose is a reducing sugar in which two α -D glucose units are joined through C_1 to C_4 linkage



Maltose contains one acetal and one hemiacetal.

Q5

The structure of threonine is

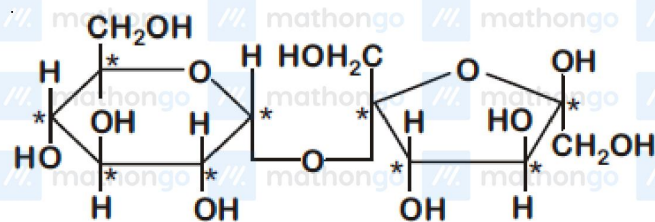


No. of chiral centres present in it = 2

Q6

Tyrosine is not an essential amino acid.

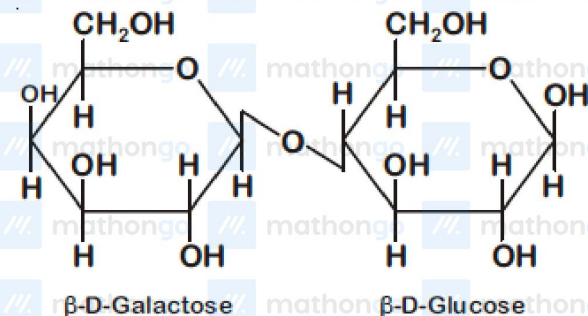
Q7



No. of chiral centres = 9

Q8

Lactose contains β -glycosidic linkage between C_1 of galactose and C_4 of glucose.



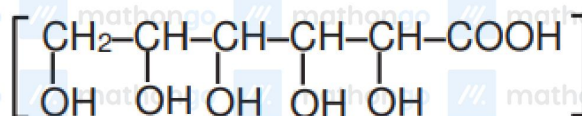
Q9

Vitamins	Deficiency Diseases
Vitamin B ₁ (Thiamine)	Beri Beri
Vitamin B ₂ (Riboflavin)	Cheilosis
Vitamin B ₆ (Pyridoxine)	Convulsions
Vitamin C (Ascorbic acid)	Scurvy

Q10

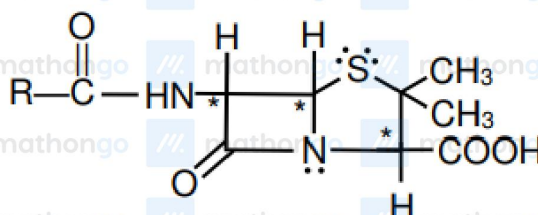
Gluconic acid

is obtained by partial oxidation of glucose by Tollen's reagent or Fehling solution or Br_2H_2O



Gluconic acid can not form hemiacetal or acetal

Q11



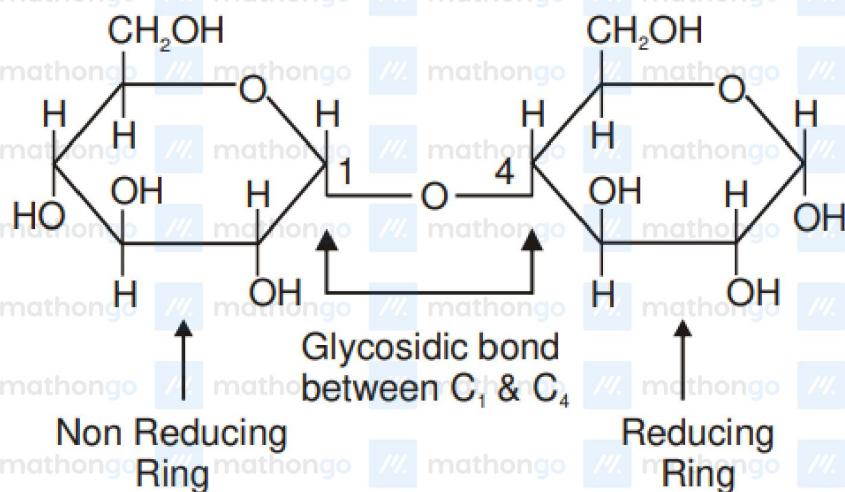
Star marked atoms are chiral centers.

Q12

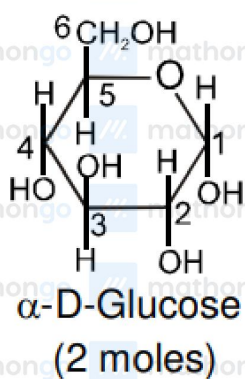
Open chain form of glucose is very small, hence does not give Schiff's test.

Q13

Maltose on hydrolysis gives 2 moles of α -D-glucose.



Hydrolysis



Lab tests	Molisch's test	Barfoed test	Biuret test
Given by	Lactose, Glucose, Fructose	Glucose	Albumin
Do not given by		Fructose	Alanine

Q15

Primary amine react with $\text{CHCl}_3 + \text{KOH}$ to give isocyanide Adenine and lysine can react with $\text{CHCl}_3 + \text{KOH}$ as they contain $-\text{NH}_2$ group.

